
VespriX
By BeatBox

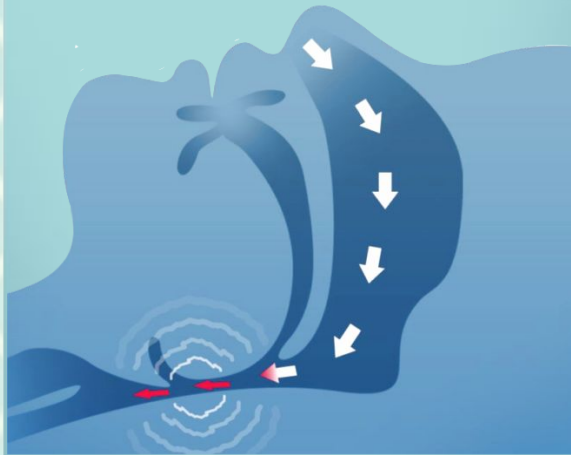
Problem

Diagnosing Sleep Apnea

While doctors and patients prefer convenient Home Sleep Apnea Tests over long lab waitlists, current tools rely on noisy data and biases towards severe cases. As a result, subtle hypopneas are missed and mild-moderate patients go undiagnosed

Current Products

- HSAT Compromise
  - Noisy, low-quality data
- Bias towards Severe AHI



PROJECT OVERVIEW

VespriX

- **Photoplethysmography (PPG)**
  - Conversion of Photocurrent to Absorbance
  - Active Filtering for High and Low Frequency Noise
  - Dual Biomarker Extraction
- **AI Detection Model**
  - Training from Clinical Datasets
  - Capture of Apneas & Hypopneas
- **Web Design & Physical Prototype**
  - Designing an Easy to use Product

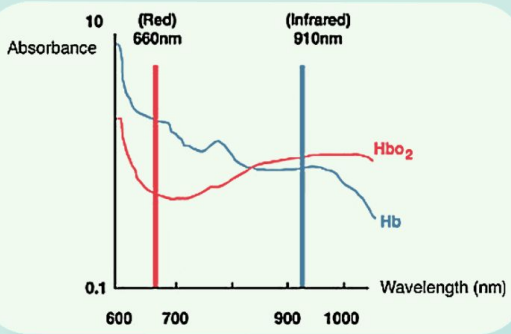
Why PPG?

- Highest ratio of clinical insight to hardware simplicity

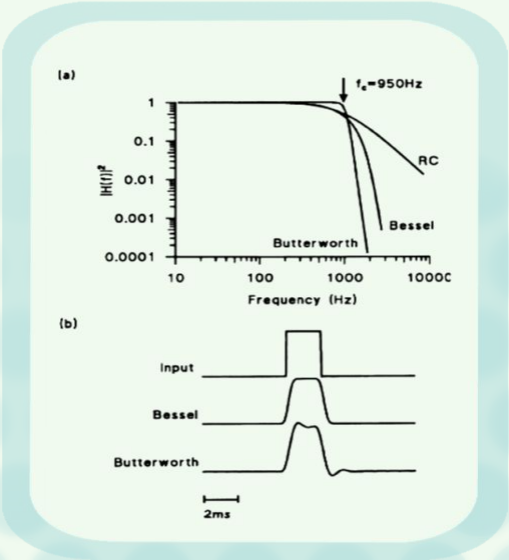
- Multi-System Signal Density
  - Captures a variety of signals (SpO_2, bpm, etc)
- Maximum Patient Comfort
  - Replaces bulky mackinges with non invasive finger sensors
- Major Cost Reduction
  - Some HSATs are disposable hardware, so costs can get high

PPG Pipeline

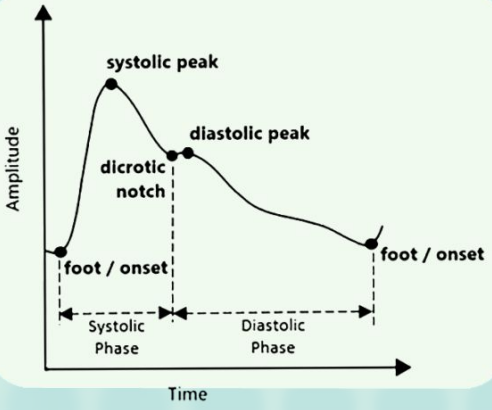
1. Conversion of R/Ir into Absorbance



2. Filtering High and Low Frequency Noise

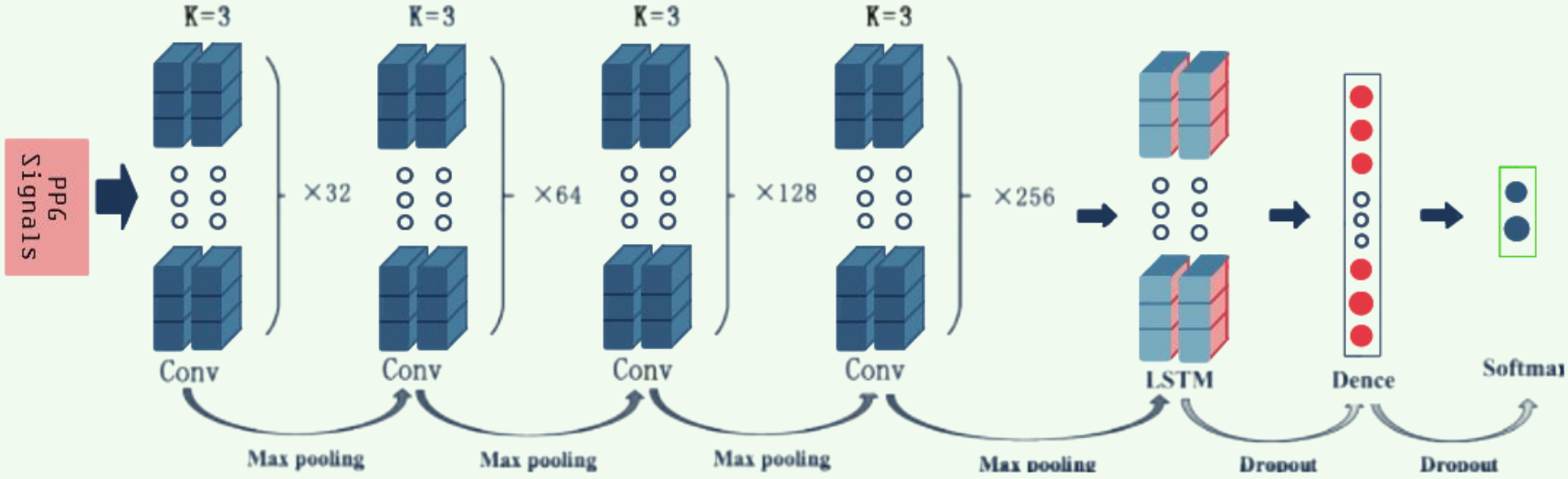


3. Feature Extraction (SpO2/Heart Rate)

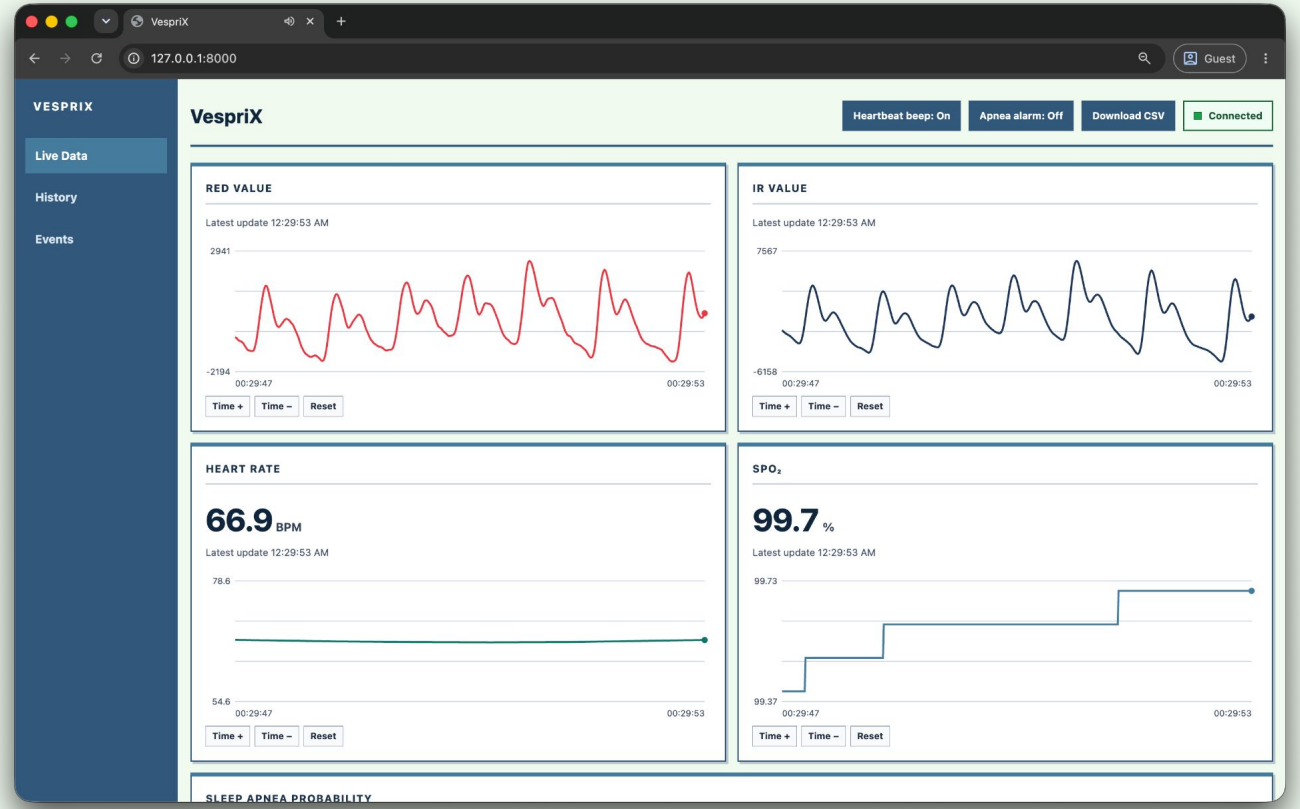


AI OVERVIEW

AUROC Rating of 85%!!



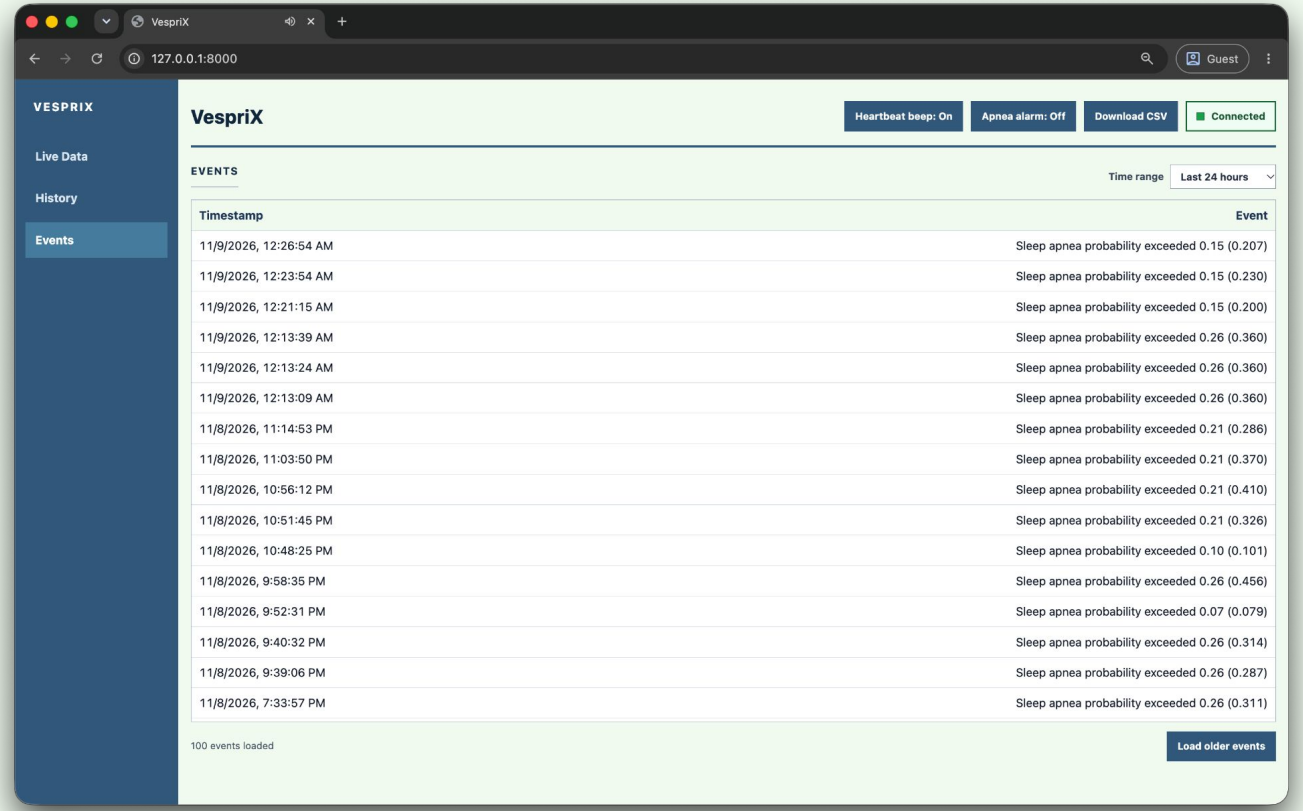
Web Browser



REAL-TIME

- Heart-Rate (bpm)
- Blood Oxygen (SpO_2)
- Beaps at Systolic Peak!
- Apnea/ Hypopnea Event Tracker

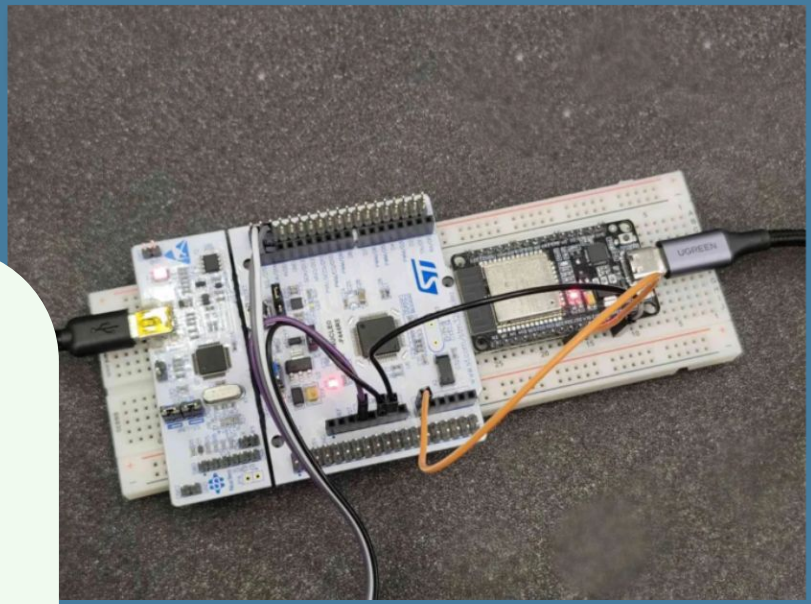
Web Browser



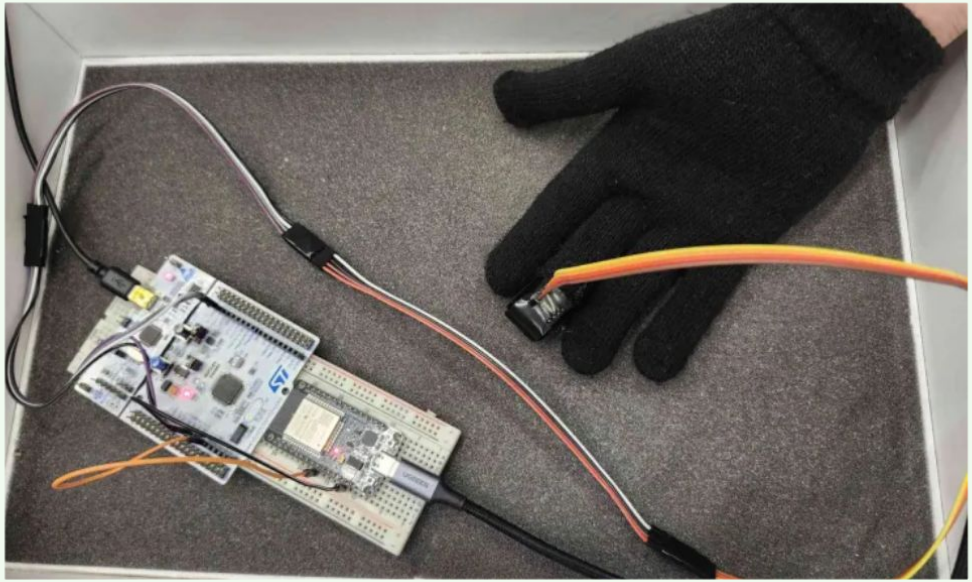
HISTORY & EVENTS

- Keeps Data & Timestamps Events
- Frequently Updated Timeline

The Physical Prototype



Demo



PROJECT SUMMARY

VespriX

- **Photoplethysmography (PPG)**
  - PPG signals can be expanded on to screen for a myriad of health topics (eg. Heart conditions)
  - Filters allow for more reliable data collection
  - Allows us to track bpm & SpO_2
- **AI Detection Model**
  - The model was able to accurately capture both Apnea and Hypopnea events
- **Web Design & Physical Prototype**
  - The physical prototype allowed for better data collection attached


Thank You





Rasheed



Alex



VespriX

By BeatBox

Caleb

Jess